

ABINASH SINHA

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SKILLS

Languages/AI Tools: Python (PyTorch, Spark, Airflow, LangChain, LangFuse), Java, SQL, Bash, Claude Code, Gemini
Frameworks/DBs: BigQuery, Spanner Graph, Docker, Git, GCP, Redis, MongoDB, Elasticsearch, Databuck, Galileo

EMPLOYMENT - 8+ yrs

Sr. Data Scientist (Autonomous Network) - Verizon | Infinite Computer Solutions | Texas Feb'25 - Present

- Architected and deployed end-to-end data wrangling and machine learning pipeline for **near real-time anomaly detection** across 100Ks of distributed telco network elements on PB scale telemetry data, delivering **~50% faster detection** and **~70% Hit Rate**
- Engineered **Agentic AI automated triaging system** utilizing SOTA LLMs (Gemini) to autonomously isolate, flag, and resolve micro-severe operational outages across complex distributed systems; Dataproc, GCS
- Designed **data-driven risk-aware network change management** converting raw system and performance metadata for actionable risk mitigation across interconnected telco network elements to reduce network outages rate; BigQuery
- Leveraged XGBoost modeling to extract hidden insights from, and textually explain extremities in **energy consumption patterns in radio access network**, championing data-driven decision-making for executive leadership; BigQuery, Dataproc

Data Scientist (Audit & Compliance) - Bank of America | Experis | Texas Aug'23 - Feb'25

- Pioneered automated text manipulation and metadata content extraction pipeline using Transformers-based architectures to ingest and categorize 10Ks of complex regulatory audit requests; **~70% reduction in document processing lead time**
- Authored comprehensive model risk development documentation and engineered ongoing automated statistical monitoring to ensure strict alignment with **Federal Reserve Board's SR 11-7 Model Risk Management framework**; used χ^2 and *KS* tests
- Developed end-to-end eXplainable AI (XAI) framework utilizing Shapley values to interpret complex NLP predictions, successfully providing clear, evidence-based recommendations and transparency to **non-technical internal auditors**
- Formulated Design of Experiments (DOE) framework to mathematically test and validate statistically significant gains in hybrid search engine relevance metrics (**NDCG@10 and Hit Rate@10**) using **BM25 and ELSER**

Senior Data Scientist (Decision Engine) - CGI | Texas Aug'22 - May'23

- Led and mentored cross-functional team of 10 subject matter experts (SMEs) and 2 junior machine learning engineers to build, test, and scale production-grade ML inference backends on cloud infrastructure.
- Developed large-scale BERT-based text manipulation and topic modeling pipeline to extract structural patterns and content understanding from 100Ks of unstructured corporate survey texts, **decreasing processing lead time by ~60%**

Data Scientist II (Search Engine) - Michaels Stores | Texas Oct'21 - Aug'22

- Pioneered large-scale multimodal data extraction pipeline, leveraging Dataproc, Cloud Run, Airflow, and Apache Spark to generate deep embeddings for 1M+ product images using to facilitate modeling for enhanced search relevance enhancement
- Built and productionized high-impact Vision-Language proof-of-concept utilizing Vision Transformers (**ViT**) and **CLIP** to evaluate image-to-text similarity, directly optimizing data quality components and content understanding
- Enhanced core algorithmic query understanding features within a distributed search architecture, improving pipeline data throughput and reducing system latency by **~2%**.

Graduate Research/Teaching Assistant (Part-Time) - University of Minnesota | Minnesota May'19 - Dec'20

- Built pipeline using RNN and time-series Transformer to classify **LSST** astronomical light curves; **DeepPhotAstro**
- Assisted with teaching 2 courses - **Algorithms, Data Structures and Machine Learning** for **~240** and **~80** students
- Devised end-to-end pipeline using convolutional LSTM neural network to **detect violence in surveillance videos**

Senior Software Engineer (Innovation Labs) - ICICI Bank | India Mar'18 - Jul'18

- Assisted with design and development of application back-end for serving Unified Payment Interface (**UPI**) transactions

Applications Developer II (Banking) - Oracle Corporation | India Aug'15 - Mar'18

- Spearheaded by leading 4 junior developers for development of flagship overdraft line-of-credit billing feature in app, **OBP**;
 - Formulated **ETL** framework for interest-rate change functionality; **decreased batch processing duration by ~7%**
 - Developed **MVP version of fee reassessment functionality** to serve loan, checking, savings and term customers
 - Assisted with initial UAT that led to eventually serving **~6 million bank accounts** in one of Tier 1 US-based bank

EDUCATION

New England College (NEC) | *M.S. - Information Technology and Cybersecurity* | **GPA 3.97/4.00** Sep'24 - Apr'26

Built time-series network intrusion detection on UNSW-NB15 and NetFlow data using self-supervision; **S3Rec** and **PatchTST**

University of Minnesota, Twin Cities (UMN) | *M.S. - Computer Science* | Research/Teaching Assistant Sep'18 - Sep'21

Plan B Project (defended Dec'23): **MOOCRec**: Mutual Info. Maximization Self-Supervision for Recommending Topic in MOOCs
- **Increased NDCG@5 and Hit Rate@5** by **~65%** and **~54%** (baseline - w/o self-supervision) MOOCube dataset

Indian Institute of Technology (IIT), Kharagpur | B.Tech.(Hons.) - **Instrumentation Engg.** Jul'11 - Jun'15

General Secretary, (Sports and Games) - Achieved 2nd position in General Championship, Sports (Best Overall Standing since 2007)